Computers in Human Behavior 104 (2020) 106159

Contents lists available at ScienceDirect

Computers in Human Behavior

journal homepage: http://www.elsevier.com/locate/comphumbeh

Do you filter who you are?: Excessive self-presentation, social cues, and user evaluations of Instagram selfies

Seoyeon Hong [a,*], Mi R. Jahng [b], Namyeon Lee [c], Kevin R. Wise [d]

[a] Rowan University, College of Communication and Creative Arts, 301 High Street, Glassboro, NJ, 08028, United States
[b] Wayne State University, Department of Communication, 525 Manoogian Hall, Detroit, MI, 48201, United States
[c] University of Missouri, School of Journalism, Water Williams Hall, Columbia, MO, 65211, United States
[d] University of Illinois at Urbana-Champaign, College of Media, 810 S. Wright Street, Urbana, IL, 61801, United States

A B S T R A C T

Innovation in the areas of social media, mobile devices, and wireless connectivity fosters new reflections on communication research, specifically in the area of self-presentation. In this paper, selfies publicly posted on Instagram (N = 1873) were analyzed to see if excessive self-presentation, operationalized as the use of photo filters in selfies, is negatively related to social media users' evaluation of the person in the selfie. The data showed that using photo filters in selfies was associated with fewer likes received from other social media users. In addition, use of social cues in selfies is positively associated with higher number of likes on Instagram. Theoretical and practical implications of these phenomena are also discussed.

1. Introduction

Social media contents are primarily based on users' voluntary sharing of information and interaction with other users. The selfie, photograph individuals take of themselves usually with a smartphone or webcam to be shared on social media, is one example of such information sharing. Selfies are forms of self-representation shared as the best photographic depiction of users (Çadırcı & Güngör, 2016; Enli & Thumim, 2012).

In 2013, "selfie" was registered as a new word by the Oxford dictionary. It is defined as "a photograph that one has taken of oneself, typically one taken with a smartphone or webcam and shared via social media" (OxfordDictionaries.com). Sorokowski and colleagues (2015) defined the selfie as "a self portrait photograph of oneself (or of oneself and other people), taken with a camera or a camera phone held at arm's length or pointed at a mirror, that is usually shared through social media (p. 124)." As the selfie is a visual depiction of self that can also send messages of how the selfie-taker is feeling, selfie posting is recognized as a normative behavior practiced by the majority of social media users (Barry et al., 2017). As selfies are considered among the most common forms of self-representation in social networking sites (Çadırcı & Güngör, 2016), studies of selfies have started from Facebook and have recently started to include Instagram, a visually oriented SNS platform (Bakhshi, Shamma, & Gilbert, 2014).

The focus of this study is to examine some of the ways in which people present themselves via selfies that are publicly posted on Instagram. Instagram, which is now one of the most popular social media platforms, with 800 million monthly and 500 million daily users (Statista.com 2017), is specifically dedicated to image based self-presentations of users via photos. While scholars have examined the social or psychological motives (Fox & Rooney, 2015; Re, Wang, He, & Rule, 2016; Sorokowska et al., 2016) or demographic differences in posting selfies on Instagram (Jang, Han, Shin, & Lee, 2015), little research has been done to explore whether certain features of selfies that draw more attention and approval from other users.

This study attempts to explain how people evaluate selfies on Instagram. To this end, publicly posted selfies on Instagram were analyzed to examine whether additional layers of social cues are related to the number of likes received by other Instagram users. As the number of 'likes' on Instagram can be considered as the rate of approval or positive evaluations (Lee & Sung, 2016), this study will assess whether higher levels of social cues in Instagram selfies are related to the number of likes from other users.

1.1. Understanding selfies as self-presentation

During face-to-face encounters, the way individuals observe each other can influence how they perceive the overall interaction. Social identity theory (Tajfel & Turner, 1986) suggests individuals strive to achieve a positive social identity in communication. Among the various

* Corresponding author.
E-mail addresses: hongs@rowan.edu (S. Hong), jahng@wayne.edu (M.R. Jahng), namyeonlee@mail.missouri.edu (N. Lee), krwise@illinois.edu (K.R. Wise).

https://doi.org/10.1016/j.chb.2019.106159
Received 16 April 2018; Received in revised form 20 June 2019; Accepted 6 October 2019
Available online 16 October 2019
0747-5632/© 2019 Elsevier Ltd. All rights reserved.

forms of posting self presentation, selfies display a positive identity by providing attractive self-image with the intention of seeking admiration from others (Sung, Lee, Kim, & Choi, 2016). The majority of selfies are taken in a way to present the individuals in the most positive way by maximizing likability (Sanghani, 2014). Such tendencies explain why selfies can be understood as a means of self-presentation of social media users (Musil, Preglej, Ropert, Klasinc, & Babič, 2017).

Self-presentation refers to tactics used to convey one's impression by controlling disclosure of one's information (Goffman, 1959). Goffman (1959) conceptualized self-presentation as an ongoing process of information management between the expressions provided by the individuals and the expressions given off while given off are nonverbal and presumably unintentional. Strategies of self-presentation involve repressing certain personal information or supplanting it to be consistent with a desired self (Kelly & McKillop, 1996). These strategies for self-presentations are frequently found among SNS users posting selfies, as many SNS sites, including Instagram, allows them to edit the photo to be the best presentation of one's appearance.

Research shows that people modify their self-presentations to be more favorable with strangers (Tice, Butler, Muraven, & Stilwell, 1995). Self-presentations in SNS can be considered as such communication with strangers, especially if posts are made public. Studies have shown that individuals who adopt online self-presentation strategies are increasing which leads to attenuation of face-to-face interaction (Papacharissi, 2002; Walker, 2000). Although self-presentations on Facebook, and Twitter have been examined (Dominick, 1999; Hong, Tandoc, Kim, Kim, & Wise, 2012; Jang et al., 2015; Papacharissi, 2002; Jensen Schau & Gilly, 2003), the visual features of selfies, as visual form of self-presentations, has not yet been examined leading to a gap in literature in selfies and self-presentations.

The act of posting selfies is a form of visual self-presentation (Re et al., 2016) as a selfie is not taken as a way to preserve memory but rather to post in a social network for others to view (Çadırcı & Güngör, 2016). Çadırcıa and Güngör asserted that uploading selfies on social media to share with others should be interpreted as an act toward the audience. In fact, Sung and colleagues (Sung et al., 2016) found that the main motivations for posting selfies are to attract attention, be acknowledged by others, and gain self-confidence from others' reaction. Similarly, Jang et al. (2015) found although teens post less selfies than adults, they are more likely to only post selfies to express emotions, more likely to focus on likes, and more likely to remove photos to be selective in self-presentations. In other words, the main purpose of posting selfies through social media is to impress others and receive attention from others through photographic presentation of self.

Omarzu (2000) suggested that there are three interpersonal goals for self-presentation: social approval, intimacy, and social control. For social media users, obtaining social approval from other users are likely to be the main motivation for posting selfies (Attrill, 2015). This explains the anticipation of receiving a 'like' which is an indicator of perceived endorsement or popularity from other social network users (Lee & Sung, 2016). It provides a way to understand the characteristics and motivations of how people evaluate contents on social media (Wallace, Buil, de Chernatony, & Hogan, 2014). Such evaluations through the 'like' play a key role in the overall use and interactions on social media (Dumas, Maxwell-Smith, Davis, & Giulietti, 2017; Lay & Ferwerda, 2018).

1.2. Excessive self-presentation

People often produce exaggerated identities of themselves to create positive images in computer-mediated communication environments (Hancock & Dunham, 2001). The main structure of self-presentation is to include a manipulation of signs (Wiley, 1994) or an embodied representation (Brewer, 1993) to impart identity. Individuals who post selfies are perceived having some level of persuasive intent for other users to view only the most desirable self image. For such persuasive intent and the technological affordances to easily change/edit photographs, selfies are often posted with additional filters or stickers to enhance one's attractiveness in physical appearances. Ma and Yang (2016) found that a majority of selfies posted on Chinese social media websites included modifications of appearances. Such findings imply how selfies are more likely a presentation for "ideal-selves" rather than simple self-presentation. Yet, no matter how hard people try to idealize, these efforts would be effective only if they are accepted by others. Self-presentation is successful when beneficiality meets believability (Schlenker, 1980). Such believability is mainly influenced by the perceived ambiguity of the presentation. If the more ambiguous the contents are, the more self-aggrandizing a self-presenter would be (Schlenker, 1975). Such pattern could evoke condemnation because others can see that they are deliberately misrepresenting themselves (Schlenker & Leary, 1982).

In this regard, excessive presentations of 'ideal' selves may decrease the popularity of selfies. DeAndreas and Walther (2011) stated that inconsistent self presentation could be perceived as more intentionally misleading because people naturally differentiate between intentional and unintentional behavior when making explanations (Malle, 2004). Excessive self-presentations like filtered photos could create such a discrepancy between the original image of oneself and a more ideally packaged image. To illustrate, producing filtered selfies may result in insufficient impression management.

Intentional persuasive effort is also known as a negative agent on individual's judgement (Boush, Friestad, & Rose, 1994; Kirmani & Campbell, 2004; Koch & Zerback, 2013). Koch and Zerback (2013) explained that when people perceive persuasive intention, it triggers reactance, which in turn attenuates participants' trust in the source and leads to a negative evaluation towards presented messages. When social media users perceive too much persuasive intent from the selfies via excessive color changes or use of filters, this would result in negative evaluations. As many social media users may be aware of how much social media users are likely to manipulate their self-presentation to fit the socially desirable images (Bazarova, Taft, Choi, & Cosley, 2013), any excessive modifications of the selfie can be perceived as misleading, thus leading to negative impressions. Thus, in an attempt to explicate this concept, the present research examines how excessive self-presentation are evaluated.

A study led by Bakhshi et al. at Yahoo Lab (2015) showed that photo filters are used to make their photos look good, fun, unique, and special. Filters are tools that enable selfie takers to enhance their photos, without the help from professional software. In general, photo filters modify colors, change light exposure or add overlay stickers in the image. Those include color changes, flower crowns, dog or cat's ears-nose-tongue, or big sparkling big eyes make selfie takers look better (Leclercq, 2016). Those filters are particularly favored by users of Instagram or Snapchat (Bakhsh, Shamma, Kennedy, & Gilbert, 2015). When users see photos filtered, they are likely aware that this is not the original content, but a modified one. As self-presentation that involves manipulating one's appearance is no longer considered authentic (Berg & Derlega, 1987), the use of filters as a manifestation of appearance manipulation demonstrates an attempt to present socially desirable self-images. This may lead to negative evaluations from other users.

Based on the previous studies in self-representation and persuasive intent, it is hypothesized that excessive self-presentation by selfie takers' will be perceived as overt persuasive intent to be evaluated only through their socially desirable self-image, not the real self. Thus, the first hypothesis tests whether selfies using excessive filters will receive more negative evaluation from other users than selfies without any filters.

H1. Selfies with filters will have fewer likes than selfies without filters.

Social cues in selfies: Theoretical explanations from Social Information Processing Theory (SIPT)

Another theoretical concept to consider in understanding self-representation and impression management on social media is the amount of social cues provided by the user in the selfies. While originally

designed to examine how people build relationships in computer-mediated settings, social information processing theory (SIPT) can provide insights to how social cues provided in selfies can relate to the favorability of the posts (Walther, 2015). SIPT posits how people utilize different communication systems, such as response time or written attitude to make up for the lack of social cues that would otherwise be available in face-to-face communication (Walther, 2011). Self-disclosure of social cues is one way users address the limitations of computer-mediated communications because it helps increase the intimacy in online relationships. Studies in SIPT consistently found how self-disclosure with increased social cues in CMC settings was related to positive relational outcomes and conversational effectiveness (Tidwell & Walther, 2002; Jang et al., 2015).

In social media environment, personal information provided on individual social media profile can serve as social cue, as it is a voluntary self-disclosure of information and indicates eagerness to communicate with others (Jang et al., 2015). Specifically for selfies, nonverbal cues can include any visual information the individuals provide about themselves as part of their selfies. For example, according to SIPT, social cues include demographic information (i.e. age, gender, socioeconomic status, residence) or personal characteristics (e.g. appearance, mood, attitude) (Bordia, 1997; Walther & Parks, 2002). Such visual representation of individuals within selfies can function as social cues (i.e. self-disclosure) that indicate eagerness to interact, a sentiment that can lead to higher audience interest and favorability.

Individuals have diverse motivations for providing social cues on their SNS profiles. Bazarova and Choi (2014) identified different motivations for posing public vs. private posts on Facebook, where social validation and self-expression/relief were the two main motivations for public statuses while relational development was the primary motivation for private messages and wall posts. Vitak and Kim (2014) identified social approval goal (i.e. seeking social acceptance from others), social control goal (i.e. engaging in social control in self-disclosure), intimacy goal (i.e. increasing and maintaining relational closeness), and personal record goal. As the main contents shared on Instagram are visuals, such as videos and photos, such motivations for self-expressions and social validations are also likely to be shared among users. For example, Instagram use has been associated with decrease in loneliness (Pittman, 2015), which Pittman and Reich (2016) suggested as the result of intimacy of visual contents in comparison to verbal contents. As the number of likes on Instagram can represent the level of support to contents (Andalibi, Ozturk, & Forte, 2017), and as self-photos tend to attract more likes and comments on Instagram (Bakhshi et al., 2014), these interactions resulting from disclosure of social-cues may explain the intimacy in Instagram in comparison to other social media platforms.

Scholars have identified several key outcomes of disclosing social cues on SNS. For example, studies have consistently identified disclosure of social cues as an important condition to positive relational outcomes and conversational effectiveness in social media environment (Hong et al., 2012; Tidwell & Walther, 2002). On social media, photos with more social cues are evaluated more positively than those with less social cues (Hong et al., 2012). Jang et al., 2015 found journalists who provide more social cues in their Twitter profiles are positively evaluated by other users. The level of warmth and intimacy with other individuals was enhanced with uses of avatars and emoticons, which is also suggested as disclosure of social cues (Cassell, Sullivan, Prevost, & Churchill, 2000). Trust to other users (Cyr, Hassanein, Head, & Ivanov, 2007) and credibility toward persuasive messages (Winter & Kramer, 2014) are also positively influenced by disclosure of social cues. In addition, the quality of responses received from others was more personable and polite when original posts included social cues of the individual in online support forums (Li, Feng, Li, & Tan, 2015).

Previous studies consistently suggest the disclosure of social cues lead to positive evaluations from other users on social media platforms. While selfies are, by definition, a visual representation of the individual, we suggest there is other personal information that can be provided within the selfie photographs, such as luxury products used by the individuals (i.e. level of wealth), physical fitness of the individuals (i.e. importance of physical fitness), or the individual's professional identity (i.e. professional uniform or job environment). As visual contents are perceived to be more intimate than verbal contents (Pittman & Reich, 2016), and self-presentations in SNS leads to more positive overall evaluations, selfies with high level of social cues are predicted to have similar levels of positive evaluations from other users, in comparison to selfies with lower social cues.

H2. Selfies with social cues will have more likes than selfies without social cues.

RQ1: Will the relationship between the number of likes and social cues in selfies be moderated by the use of filters in selfies?

2. Method

2.1. Sampling procedure

Content analysis of Instagram posts was conducted to collect the data. A total of 2030 public Instagram posts were sampled by using 'selfies' as a search term (Ma & Yang, 2016; Kalayeh, Seifu, LaLanne, & Shah, 2015). Instagram posts were cleaned in a 3-step process using the previous standard for selecting selfies (Souza et al., 2015). First, a simple random sample of public Instagram posts with hashtag 'selfies' (n = 2030) was collected by the coders using and stored as screenshots. Then researchers excluded images that did not include a person's face or were posted within an hour from data collection period, since these posts did not have ample time for the exposure to receive a legitimate number of "likes" on their posts (Carbone, 2018; Takumi, 2016). Lastly, any posts for commercial or promotional purposes were excluded. As a result, a total of 1873 selfies were selected for the main analysis. This is an appropriate sample size for the sampling from the Internet and individual communication as shown by Riff and colleagues (1998)'s examples. In their book (p.121), Internet-based content analysis included the wide range of sample size: from 64 (Paul, 2001) to 487 (Wicks & Souley, 2003).

Out of our samples, 90.7% (n = 1700) were female and 9.2% (n = 173) of them were male. Approximately half were Caucasian (50.7%, n = 951) followed by 25% (n = 468) Hispanic, 18.8% (n = 352) Asian, and 4.6% African-American. Another 0.8% (n = 15) corresponded to an "other" racial category.

2.2. Inter-coder reliability

A codebook with instructions and examples was developed from previous studies in selfies and self-presentation and social cues (Goffman, 1959; Hong et al., 2012; Tidwell & Walther, 2002; Walther, 2011). Approximately 10 percent (n = 189) of the Instagram selfies were coded independently by three coders who also participated in the main coding procedure. Three coders are enough to protect the data from coding bias as two coders are common in content analyses and meet published standards (Austin & Pinkleton, 2006). Disagreements among coders were resolved through further discussion and clarifications. Inter-coder reliability was calculated using the SPSS macro by Hayes and Krippendorf (2007) for Krippendorf's alpha. Inter-coder reliability was acceptable for all variables: gender (a = 1), race (a = 0.78); use of filter (a = 0.87); and social cues (a = 0.70).

2.3. Operational definitions & coding procedures

The unit of analysis for this study was each selfie sampled from Instagram. Every selfie was coded for the following categories: (1) number of followers for each account, (2) number of highest likes from the selfie takers' Instagram accounts, (3) gender of the person in the

selfie, (4) race of the person in the selfie, (5) use of photo filters on selfies, (6) the presence of social cues in the selfie and (7) number of likes received for the selfie.

2.4. Independent variables

Use of filters in selfies. Excessive self-presentation was coded dichotomously (y/n) for the presence of photo filters including color changes, and sticker filters, such as flower crowns, dog or cat's ears-nose-tongue, or big sparkling big eyes on selfies (Leclercq, 2016) (see Fig. 1).

Social cues. Social cues were operationalized as the additional information upon which an impression can be made (Hong et al., 2012; Milyavskaya, Reoch, & Koestner losier, 2010). In the current study, selfies taken showing luxury goods, fitness center as a background, athletic clothes, or professional identity (e.g., pilot, flight attendant, doctor) were coded as selfies with social cues. When no other social cues were present in the selfie, it was coded as none (see Fig. 2).

2.5. Dependent variable

Number of Likes. Consistent with previous research, the number of 'likes' a selfie have received was used in determining how other Instagram users evaluated the selfie (Wallace et al., 2014). To avoid violating normal distribution assumption, the number of likes was normalized in percent (%) suggested by Pearson (1895) and others (Wiebe et al., 2006; Wilk & Gnanadesikan, 1968). Accordingly, the number of like was divided by the highest number of like from selfie takers' instagram account in a way to change raw data with high standard variation into notionally common scale. The unit of analysis remained the same.

3. Results

H1 predicted the difference in the number of likes between selfies using filters compared to those without filters. This hypothesis was tested with an independent sample *t*-test. Results showed statistically significant differences in the number of likes between selfies using photo filters in comparison to selfies not using photo filters, $t(1873) = 8.29$, $p < .001$. Selfies using filters had significantly lower number of likes ($M = 41.39$, $SD = 21.12$) in comparison to selfies without any filters ($M = 53.49$, $SD = 27.59$). Therefore, H1 was supported.



Fig. 1. An example of selfie using photo filter.



Fig. 2. An example of selfie using social cue.

H2 predicted that using social cues would be positively related to the number of likes received from other users. An independent samples *t*-test was conducted to test this hypothesis. There was a statistically significant difference in the number of likes between selfies with social cues compared to selfies without social cues $t\ (1872) = 7.248$, $p < .001$. Specifically, selfies having social cues had higher number of likes ($M = 64.57$, $SD = 32.08$) in comparison to selfies without social cues ($M = 49.39$, $SD = 25.74$). Therefore, H2 is supported.

For both hypotheses, the t-tests were replicated with the original number of likes, i.e. the non-standardized number of likes, $SD = 2000.46$. and produced the same results. For H1, t(1872) score was $= 4.145$, $p < .001$. For H2, t(179) score was 3.246, $p < .001$) when equal variance is not assumed (see Table 1 for the detail).

Lastly, RQ1 assessed the moderating role of social cues on the number of likes for each selfie. According to the results of two-way analysis of variance, the use of filters in selfies did not moderate the relationship between social cues in selfies and the number of likes received, $F(1, 1873) = 0.005$, $p = .944$.

4. Discussion

The purpose of the current study was to offer a foundation for theoretical discussion among scholars from diverse fields and to provide a baseline understanding of selfies, as a new phenomenon on social media. This study examined what features of selfies attracts other users to provide positive feedback on Instagram. Using self-presentation and social information processing theory as the main theoretical background, this study hypothesized that 1) use of filters on selfies, operationalized as excessive self-presentation, would be negatively correlated with the number of likes received and 2) the presentation of self through social cues would have more number of likes. Both hypotheses were confirmed from the content analyses of selfies posted on Instagram. Results indicated that selfies using filters had less number of likes compared to selfies without any filters. On the other hand, when selfies included social cues of the selfie-taker, it had more likes in

Table 1
T-test results for use of filter and social cues on social media user evaluation.

		Number of likes		T-Value	Non standardized Number of likes		T-value
		M	SD		M	SD	
Filter	Absence	53.49	27.59	8.29***	968.14	3774.47	7.887***
	Presence	41.39	21.12		164.79	498.90	
Social Cue	Presence	64.57	32.08	7.248***	2409	25.74	3.25***
	Absence	49.39	25.74		622.49	2596.35	

*p < .05, **p < .01, ***p < .001.

comparison to selfies without any social cues of the individuals.

The current findings suggest a meaningful reconsideration of examining selfies in terms of impression management and self-presentations among SNS users. Theoretically, this study introduces the features of selfies, social cues in the photos, to be more predictive of positive evaluations and approval from other users. Although previous studies examined selfie itself as a form of self-presentations (Ma and Yang, 2016; Sorokowska et al., 2016), the results indicate selfies providing additional information about the individuals in the form of social cues, such as professional identity or status of wealth, may be perceived as further intention to engage with other social media users rendering more likes. According to the Social Information Processing Theory, additional information provided in computer-mediated settings express willingness to engage with others, leading to positive impressions of the interactions and participants of the interactions (Tidwell & Walther, 2002). This study suggests that even with the selfie, which is already a form of visual self-presentation, adding additional visual information about the self can be interpreted as additional social cues that are perceived positively by other users.

On the other hand, the number of likes was lower for selfies posted with filters, such as stickers or excessive use of color filters compared to selfies without such filters. This result was interpreted in light of excessive intent in self-presentations and impression management. Studies in self-presentations suggest excessive excessive intent toward others to only think positively of an individual's self-image may trigger reactance to engage in SNS environment (Koch & Zerback, 2013). The findings from the current study confirm that excessive intents to present the most ideal self image are perceived negatively by other users. For example, Re et al. (2016) found the prevalence of self-favoring biases among selfie takers. While selfie takers may regard photo filters to be increasing the visual attractiveness of themselves, this can be perceived by other users as an ingenuine behavior because it reflects the intent to only present the ideal depiction of self.

Narcissistic motivations are found to be a significant predictor for selfie posting behavior (Etgar & Amichai-Hamburger, 2017; Fox & Rooney, 2015; Sorokowski et al., 2015; Weiser, 2015). Additionally, people perceived people appeared in selfies are more narcissistic in comparison to people in photos taken by others (Kramer et al., 2017). Re et al. (2016) argue that narcissism is the cause why people evaluate people selfies less attractive. However, aforementioned studies were comparing selfie and other photos in general and ignored the possibility of multi-faceted concepts that compose selfie. In this regard, this finding is particularly interesting as we only collected selfies to see how such evaluation are formatted by investigating different construction of selfies. It is reasonable to assume that selfie takers who possess relatively high narcissistic tendency might be the one using filters more. Future studies should explore whether excessive use of filters in conjunction with personality traits.

This study is meaningful in three ways. First, this paper tries to answer to the calls from scholars requesting to test both positive and negative effects of selfies in user evaluation (Kramer et al., 2017). As our data of 1873 selfies suggests, self-presentation is not always a successful way to achieve positive outcomes. Second, this is the first study using an actual number of likes received from the other users in a form of self-presentation. With this external validity, the study broadens the understanding of how social media users evaluated the contents on social media. By investigating what influences on evaluation toward selfies, this study indubitably facilitates a deeper understanding of in selfie-posting behavior among social media users. Lastly, our results suggest that selfie takers and social media users should be cautious when posting selfies as it could not be working as they intended. Using social cues in selfies are recommended, yet what types of social cues could be a strong indicator of positive evaluation remains answered.

4.1. Limitation & future directions

This study sought to explore what features of selfies make them a popular and well-liked visual content in Instagram. While this study adds to the literature in impression management and self-presentation by examining which specific features within selfies predicts the likability of the contents, there are a few limitations that can be developed to future exploration of selfie culture. First, the majority of the selfies sampled for this study were posted by female users of Instagram. While studies consistently show females are more likely to post selfies than men (Weiser, 2015), focusing primarily in male-posted selfies can provide different features of the selfies to be attractive to SNS users increasing the number of likes. Additionally, because the number of likes can constantly change, the samples used in this study was limited to the specific timeframe chosen to collect the Instagram posts. To resolve this issue we did not include selfies that were posted within 1 h from the selfie was posted. However, this 1 h time frame was suggested by a trade publication (Carbone, 2018), not a scholarly article. Finally, because this study primarily focused on describing the selfies posted Instagram, any individual characteristics of users, such as personality or familiarity in posting selfies, and the effects of these characteristics on evaluations of selfies could not be explained with these data. Future studies examining behavioral and psychological factors that can impact how individuals evaluate different selfies can provide insights into the motivations and emotions explaining how and why SNS users like or dislike selfies they see on Instagram.

5. Conclusion

Overall, this study represents a preliminary attempt to explore the very common use of filters to enhance self-presentation online, and how this use relates to common indicators of social status. Theoretical ramifications include a conceptualization of filter use as excessive self-presentation. Practical applications include a greater understanding of how people currently use the affordances (e.g. filters) in new applications (e.g. Instagram) to influence others' perceptions.

References

Andalibi, N., Öztürk, P., & Forte, A. (2017). February). Sensitive self-disclosures, responses, and social support on instagram: The case of# depression. In *Proceedings of the 17th ACM conference on computer supported cooperative work & social computing* (pp. 1485–1500).

Bakhshi, S., Shamma, D. A., & Gilbert, E. (2014, April). Faces engage us: Photos with faces attract more likes and comments on instagram. In *Proceedings of the 32nd annual ACM conference on Human factors in computing systems* (pp. 965–974). ACM.

Bazarova, N. N., & Choi, Y. H. (2014). Self-disclosure in social media: Extending the functional approach to disclosure motivations and characteristics on social network sites. *Journal of Communication, 64*(4), 635–657.
Bakhshi, S., Shamma, D. A., Kennedy, L., & Gilbert, E. (2015, April). Why we filter our photos and how it impacts engagement. In *Ninth International AAAI Conference on Web and Social Media*.
Barry, C. T., Doucette, H., Loflin, D. C., Rivera-Hudson, N., & Herrington, L. L. (2017). "Let me take a selfie": Associations between self-photography, narcissism, and self-esteem. *Psychology of Popular Media Culture, 6*(1), 48.
Bazarova, N. N., Taft, J. G., Choi, Y. H., & Cosley, D. (2013). Managing impressions and relationships on Facebook: Self-presentational and relational concerns revealed through the analysis of language style. *Journal of Language and Social Psychology, 32*(2), 121–141.
Berg, J. H., & Derlega, V. J. (Eds.). (1987). *Self-disclosure: Theory, research, and therapy*. Plenum.
Bordia, P. (1997). Face-to-face versus computer-mediated communication: A synthesis of the experimental literature. *Journal of Business Communication, 34*(1), 99–118.
Boush, D. M., Friestad, M., & Rose, G. M. (1994). Adolescent skepticism toward TV advertising and knowledge of advertiser tactics. *Journal of Consumer Research, 21*(1), 165–175.
Brewer, M. B. (1993). Social identity, distinctiveness, and in-group homogeneity. *Social Cognition, 11*(1), 150–164.
Çadırcı, O. T., & Güngör, S. A. (2016). Love my selfie: Selfies in managing impressions on social networks. *Journal of Marketing Communications*, 1–20.
Carbone, L. (2018, April 1). How often should you post to instagram?. Retrieved from https://later.com/blog/how-often-post-to-instagram/.
Cassell, J., Sullivan, J., Prevost, S., & Churchill, E. (Eds.). (2000). *Embodied conversational agents*. Cambridge, MA: The MIT Press.
Cyr, D., Hassanein, K., Head, M., & Ivanov, A. (2007). The role of social presence in establishing loyalty in e-service environments. *Interacting with Computers, 19*(1), 43–56.
Dumas, T. M., Maxwell-Smith, M., Davis, J. P., & Giulietti, P. A. (2017). Lying or longing for likes? Narcissism, peer belonging, loneliness and normative versus deceptive like-seeking on instagram in emerging adulthood. *Computers in Human Behavior, 71*, 1–10.
Enli, G. S., & Thumim, N. (2012). Socializing and self-reprtesentation online: Exploring Facebook. *Observatorio (OBS*) Journal, 6*(1), 87–105.
Etgar, S., & Amichai-Hamburger, Y. (2017). Not all selfies took alike: Distinct selfie motivations are related to different personality characteristics. *Frontiers in Psychology, 8*, 842.
Fox, J., & Rooney, M. C. (2015). The dark triad and trait self-objectification as predictors of men's use and self-presentation behaviors on social networking sites. *Personality and Individual Differences, 76*, 161–165.
Goffman, E. (1959). *The presentation of self in everyday life*. Garden City, NY: Doubleday-Anchor.
Hancock, J. T., & Dunham, P. J. (2001). Impression formation in computer-mediated communication revisited: An analysis of the breadth and intensity of impressions. *Communication Research, 28*(3), 325–347.
Hayes, A. F., & Krippendorf, K. (2007). Answering the call for a standard reliability measure for coding data. *Communication Methods and Measures, 1*, 77–89.
Hong, S., Tandoc, E., Kim, E. A., Kim, B., & Wise, K. (2012). The real you? The role of visual cues and comment congruence in perceptions of social attractiveness from Facebook profiles. *Cyberpsychology, Behavior, and Social Networking, 15*(7), 339–344.
Jang, J. Y., Han, K., Shih, P. C., & Lee, D. (2015). *Generation like: Comparative characteristics in instagram. ACM conference on human factors in computing systems (CHI)*. https://doi.org/10.1145/2702123.2702555.
Jensen Schau, H., & Gilly, M. C. (2003). We are what we post? Self-presentation in personal web space. *Journal of Consumer Research, 30*(3), 385–404.
Krämer, N. C., Feurstein, M., Kluck, J. P., Meier, Y., Rother, M., & Winter, S. (2017). Beware of selfies: The impact of photo type on impression formation based on social networking profiles. *Frontiers in Psychology, 8*, 188.
Kirmani, A., & Campbell, M. C. (2004). Goal seeker and persuasion sentry: How consumer targets respond to interpersonal marketing persuasion. *Journal of Consumer Research, 31*(3), 573–582.
Kelly, A. E., & McKillop, K. J. (1996). Consequences of revealing personal secrets. *Psychological Bulletin, 120*(3), 450–465.
Koch, T., & Zerback, T. (2013). Helpful or harmful? How frequent repetition affects perceived statement credibility. *Journal of Communication, 63*(6), 993–1010.
Lay, A., & Ferwerda, B. (2018). Predicting users' personality based on their 'liked' images on instagram. In *The 23rd International on Intelligent User Interfaces*. March 7-11, 2018.
Leclercq, C. (2016). Do you "LensIt"? A call for research on modified selfies. Retrieved from https://mastersofmedia.hum.uva.nl/blog/2016/10/20/do-you-lensit-a-call-for-research-on-modified-selfies/.
Lee, J. A., & Sung, Y. (2016). Hide-and-seek: Narcissism and "Selfie"-related behavior. *Cyberpsychology, Behavior, and Social Networking, 19*(5), 347–351. https://doi.org/10.1089/cyber.2015.0486.
Li, S., Feng, B., Li, N., & Tan, X. (2015). How social context cues in online support-seeking influence self-disclosure in support provision. *Communication Quarterly, 63*(5), 586–602.
Ma, J., & Yang, Y. (2016). What can we know from selfies-An exploratory study on selfie and the implication for marketers. In *2016 global marketing conference at Hong Kong proceedings* (pp. 597–601). Global Alliance of Marketing & Management Associations.
Malle, B. F. (2004). *How the mind explains behavior. Folk Explanation, Meaning and Social Interaction*. Massachusetts: MIT-Press.
Musil, B., Preglej, A., Ropert, T., Klasinc, L., & Babič, N.Č. (2017). What is seen is who you are: Are cues in selfie pictures related to personality characteristics? *Frontiers in Psychology, 8*, 82.
Omarzu. (2000). A disclosure decision model: Determining how and when individuals will self- disclose. *Personality and Social Psychology Review, 4*(2), 174–185.
Papacharissi, Z. (2002). The presentation of self in virtual life: Characteristics of personal home pages. *Journalism & Mass Communication Quarterly, 79*(3), 643–660.
Paul, M. J. (2001). Interactive disaster communication on the Internet: A content analysis of sixty-four disaster relief home pages. *Journalism & Mass Communication Quarterly, 78*, 739–753.
Pittman, M. (2015). Creating, consuming, and connecting: Examining the relationship between social media engagement and loneliness. *The Journal of Social Media in Society, 4*(1).
Pittman, M., & Reich, B. (2016). Social media and loneliness: Why an Instagram picture may be worth more than a thousand Twitter words. *Computers in Human Behavior, 62*, 155–167.
Re, D. E., Wang, S. A., He, J. C., & Rule, N. O. (2016). Selfie-indulgence: Self-favoring biases in perception of selfies. *Social Psychological and Personality Science, 7*(6), 588–596.
Riff, D., Lacy, S., Fico, F., & Watson, B. (1998). *Analyzing media messages: Using quantitative content analysis in research*. Routledge.
Sanghani, R. (2014). Why we really take selfies: The "terrifying" reasons explained. The telegraph. April 11. Available online: http://www.telegraph.co.uk/women/107.
Schlenker, B. R. (1975). Self-presentation: Managing the impression of consistency when reality interferes with self-enhancement. *Journal of Personality and Social Psychology, 32*, 1030–1037.
Schlenker, B. R. (1980). *Impression management: The self-concept, social identity, and interpersonal relationships*. Monterey, CA: Brooks/Cole.
Schlenker, B. R., & Leary, M. R. (1982). Audiences' reactions to self-enhancing, self denigrating, and accurate self-presentations. *Journal of Experimental Social Psychology, 18*, 89–104.
Souza, F., de Las Casas, D., Flores, V., Youn, S., Cha, M., Quercia, D., & Almeida, V. (2015, November). Dawn of the selfie era: The whos, wheres, and hows of selfies on Instagram. In *Proceedings of the 2015 ACM on conference on online social networks* (pp. 221–231). ACM.
Sorokowski, P., Sorokowska, A., Oleszkiewicz, A., Frackowiak, T., Huk, A., & Pisanski, K. (2015). Selfie posting behaviors are associated with narcissism among men. *Personality and Individual Differences, 85*, 123–127.
Sorokowska, A., Oleszkiewicz, A., Frackowiak, T., Pisanski, K., Chmiel, A., & Sorokowski, P. (2016). Selfies and personality: Who posts self-portrait photographs? *Personality and Individual Differences, 90*, 119–123.
Sung, Y., Lee, J. A., Kim, E., & Choi, S. M. (2016). Why we post selfies: Understanding motivations for posting pictures of oneself. *Personality and Individual Differences, 97*, 260–265.
Tajfel, H., & Turner, J. C. (1986). The social identity theory of intergroup behaviour. In S. Worchel, & W. G. Austin (Eds.), *Psychology of intergroup relations*. Chicago, IL: Nelson-Hall.
Takumi. (2016, February 9). The half-life of Instagram posts. Retrieved from https://blog.takumi.com/the-half-life-of-instagram-posts-3db61fb1db75.
Tice, D. M., Butler, J. L., Muraven, M. B., & Stillwell, A. M. (1995). When modesty prevails: Differential favorability of self-presentation to friends and strangers. *Journal of Personality and Social Psychology, 69*(6), 1120–1138.
Tidwell, L. C., & Walther, J. B. (2002). Computer-mediated communication effects on disclosure, impressions, and interpersonal evaluations: Getting to know one another a bit at a time. *Human Communication Research, 28*(3), 317–348.
Vitak, J., & Kim, J. (2014). February). You can't block people offline: Examining how Facebook's affordances shape the disclosure process. In *Proceedings of the 17th ACM conference on Computer supported cooperative work & social computing* (pp. 461–474). ACM.
Wallace, E., Buil, I., de Chernatony, L., & Hogan, M. (2014). Who "likes" you… and why? A typology of Facebook fans: From "fan"-atics and self-expressives to utilitarians and authentics. *Journal of Advertising Research, 54*(1), 92–109.
Walker, K. (2000). It's difficult to hide it": The presentation of self on Internet home pages. *Qualitative Sociology, 23*(1), 99–112.
Walther, J. B. (2011). Theories of computer-mediated communication and interpersonal relations. *The handbook of interpersonal communication, 4*, 443–479.
Walther, J. B., & Parks, M. R. (2002). Cues filtered out, cues filtered in. *Handbook of interpersonal communication, 3*, 529–563.
Wicks, R. H., & Souley, B. (2003). Going negative: Candidate usage of Internet Web sites during the 2000 presidential campaign. *Journalism & Mass Communication Quarterly, 80*, 128–144.
Wiebe, N., Vandermeer, B., Platt, R. W., Klassen, T. P., Moher, D., & Barrowman, N. J. (2006). A systematic review identifies a lack of standardization in methods for handling missing variance data. *Journal of Clinical Epidemiology, 59*(4), 342–353.
Winter, S., & Krämer, N. C. (2014). *A question of credibility–Effects of source cues and recommendations on information selection on news sites and blogs*.
Wilk, M. B., & Gnanadesikan, R. (1968). Probability plotting methods for the analysis for the analysis of data. *Biometrika, 55*(1), 1–17.